

ATOMIC STRUCTURE – ELECTRONIC STRUCTURE

KEY VOCABULARY

Energy level (shell): a region around the nucleus where electrons are found.

Electronic structure: how electrons are arranged in the shells, e.g. 2,8,1.

Outer shell: the highest occupied energy level — controls reactions.

Group number: equals the number of electrons in the outer shell.

COMMON MISCONCEPTIONS

~~— The first shell holds 8 electrons.~~

✓ The first shell holds a maximum of 2.

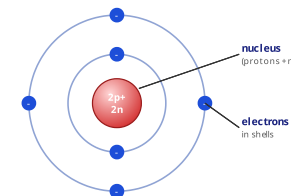
~~— Electrons fill the outer shell first.~~

✓ Inner shells fill before outer shells.

~~— Period number = outer electrons.~~

✓ Group number = outer electrons; period = number of shells.

ELECTRON SHELLS



Electrons fill shells **2, 8, 8** ... from the inside out.

1

SHELL CAPACITY

State the maximum number of electrons in the **first and second** shells. [2 marks]

2

WRITE THE STRUCTURE

Magnesium has 12 electrons. Write its **electronic structure**. [1 mark]

Answer: _____ (e.g. 2,8,...)

3

GROUP FROM STRUCTURE

An atom has the structure 2,8,7. State its **group and period**. [2 marks]

4

OUTER ELECTRONS

Explain why elements in the same group have similar reactions. [2 marks]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

The number of occupied ____ equals the ____ number of an element.

group

period

shells

two

6

DRAW AN ATOM

Oxygen is 2,6. **State** how many electrons are in its outer shell. [1 mark]
